Aaliya Alam

+1 (734) 492-6248 · aaaliya@umich.edu · LinkedIn: Aaliya Alam · GitHub: aaliya-work

Education

M.S. in Robotics, University of Michigan, Ann Arbor *[2026–Present]*

Relevant Coursework: ROB 501 (Math in Robotics), CSE 598 (Reinforcement Learning), ROB 550 (Robotics Systems Lab)

B.Tech in Computer Science and Engineering (AI and Robotics) *[2022–2026]*

Vellore Institute of Technology, Chennai | *CGPA: 8.8/10* | *3.8/4*

Experience

Robotic Perception Intern, Sakar Robotics, Pune, India | Completion Letter *[Feb 2026–July 2026]*

- Built sparse point-cloud wall reconstruction and RGB-D mapping pipelines, reworking 3D meshing with TSDF integration to achieve drift-free, sub-mm reconstruction and robust wall/obstacle extraction.
- Led perception R&D for mobile manipulators: robust spatial tracking of a multi-part foam assembly under dynamic lighting and backgrounds; resolved depth, pose-estimation and pre-grasp planning bottlenecks, cutting execution time by several seconds.

MARVO: Marine-Adaptive Radiance-aware Visual Odometry | arXiv preprint *[Oct 2025–Jan 2026]*

- Co-developed an underwater visual odometry pipeline: a physics-aware radiance adapter on a transformer-based matcher (LoFTR), followed by visual-inertial-barometric fusion in a fixed-lag factor graph (GTSAM).
- RL-based pose-graph optimization (SAC, GNN encoding) achieved 1.73 m ATE and 1.2% drift on real underwater field data.

HADVel: Hydrodynamics Aware Deep Velocity Estimation for AUV *[May 2025–May 2026]*

Final Year Research Project, VIT Chennai

- LSTM-fused velocity estimation using IMU quaternion feedback: 98.5–99.0% correlation with encoder ground truth, vs. 96.8% for raw differential-pressure channels.
- 9.4% relative MAE across multi-speed validation, with $<$50 mm/s absolute error over 0–0.6 m/s.

Data Scientist Intern, Fifthgen Tech Solutions, Noida, India | Completion Certificate *[May 2024–July 2024]*

- Built a real-time construction-site safety system (YOLOv8 detection, Mask R-CNN segmentation) for PPE compliance and hazardous-zone identification, optimized for real-time inference at 92% mAP.

Summer Research Intern, CHAIR, VIT Chennai | Completion Certificate *[May 2025–Aug 2025]*

- CycleGAN-based retinal image harmonization with a custom preservation loss to conserve microaneurysms, haemorrhages, hard exudates and optic disk; 0.93 SSIM and 33 dB PSNR on real Indian retinal data (MDRF, Vision Research Foundation).

President, Advances in Autonomous Unmanned Vehicles Club, VIT Chennai *[Aug 2024–Aug 2025]*

- Led a 48-student team; ran hands-on workshops on ROS, embedded systems and control (PID/LQR); managed budget and faculty/administration coordination.

Projects

MIRA Autonomous Underwater Vehicle, Programming Lead, Dreadnought Robotics *[Sep 2023–Feb 2024]*

2nd Place Internationally, TAU Autonomy Centre Competition, Norway 2024 | Certificate | Demo

- Dual-camera ArUco marker detection and pose estimation with sub-centimeter accuracy; ROS pipeline deployed on Jetson Xavier, enabling autonomous navigation around underwater obstacles.
- PID tuning mechanism using AdaGrad optimization with graphing; led the programming team on ML/DL-based AUV performance work.

7D Language-Biased VLA for Robot Manipulators | GitHub

- Tested an idea involving a Vision-Language-Action policy conditioning 7-D robot actions on language instructions, separating language-driven action modulation from visual state prediction; evaluated against a CLIP-based vision-only baseline with interpretable per-joint modulation.

RGB-D Perception and Odometry-Based Mapping | GitHub

- Kinect + ROS pipeline converting depth images to 3D point clouds with an online nearest-obstacle monitor issuing stop/clear advisories; odometry-based 2D/3D trajectory mapping for a differential-drive robot.

Technical Skills

Languages & Frameworks: C/C++, Python, ROS/ROS2, PyTorch, TensorFlow, OpenCV, CMake, Git, Linux
Simulation & Tools: Gazebo, RViz, AirSim, MATLAB Simulink, Foxglove, QGroundControl, CoppeliaSim